Problem 5 | Problems T5_MechanicalProperties.pdf #5 | Thermal stress · constant length

Thermal Stress in a Constrained Steel Wire: the Diameter that Cancels the Applied Stress

Original statement · Problems T5_MechanicalProperties.pdf, p. 3

5. A steel wire, initially 1 meter in length, is stretched by a force of 1400 N at a temperature of 20 ºC. After the force is applied, the wire is held at a constant length and then heated to 40 ºC. Determine the required diameter of the wire to ensure that the thermal expansion stress compensates for the applied stress.

>

> *Data: E = 207 GPa;* $\alpha_l = 12.5 \times 10^{-6}$ *ºC⁻¹;*

Official Solution: 5.86 mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, linear elastic wire.** The wire is loaded only along its axis and it stays elastic, so $\sigma = E\varepsilon$ [Session 11 Slide 13].
2. **Constant length while heating.** After the force is applied, the wire is held at a constant length: the total length, and therefore the total strain ε, does not change during the heating.
3. **Linear thermal expansion.** A temperature rise ΔT produces a free thermal strain $\varepsilon_T = \alpha_l \, \Delta T$. This law is not in the slides of Session 11 and 12 (it belongs to the thermal properties of Topic 7); it is given by the sheet through the datum α_l.
4. **Strains add.** The total strain is the elastic (mechanical) strain plus the thermal strain, $\varepsilon = \sigma/E + \alpha_l \, \Delta T$, and E and α_l are constant between 20 and 40 °C.
5. **"Compensates".** The thermal stress compensates the applied stress when the net stress of the wire after heating is zero.
6. **Circular, uniform section.** The area $A = \pi d^2/4$ is the same all along the wire and does not change with the temperature.

Data

Quantity	Symbol	Value	In SI
Force	F	1400 N	1400 N
Initial temperature, final temperature	T_1, T_2	20 °C, 40 °C	
Temperature rise	ΔT	20 °C	20 K

Quantity	Symbol	Value	In SI
Young's modulus	E	207 GPa	207×10^9 Pa = 207 000 N/mm²
Linear expansion coefficient	α_l	12.5×10^{-6} °C⁻¹	12.5×10^{-6} K⁻¹
Initial length	l_0	1 m	not needed

Unknown: the diameter d of the wire.

PHASE 2 · FRAMES & STRATEGY

There is a single direction, the axis of the wire; no change of basis is involved.

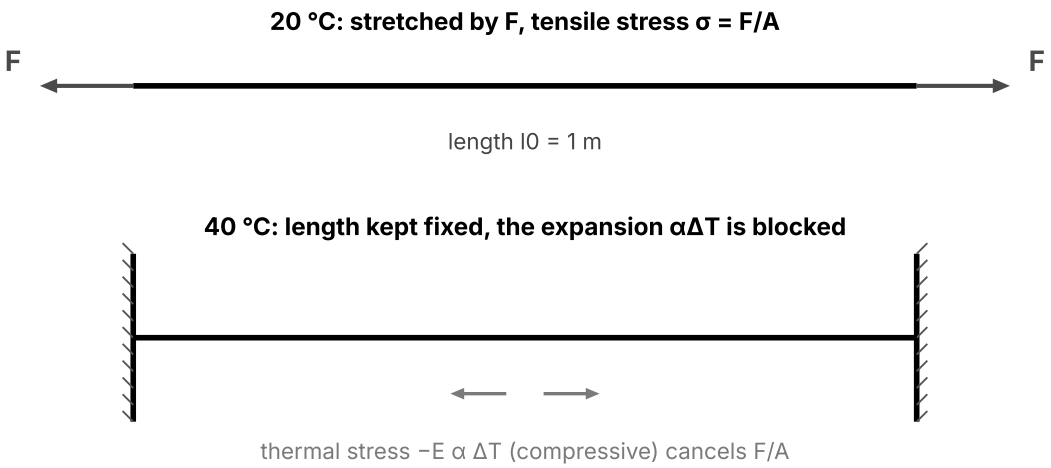


Figure 5. Problem 5: the wire is stretched at 20 °C and then held at constant length while it is heated to 40 °C (schematic).

Strategy. The key is to follow the strain, because the length is the quantity that is fixed.

1. At 20 °C the force produces the stress $\sigma_F = F/A$ and the strain $\varepsilon_1 = \sigma_F/E$.
2. At 40 °C the strain is the same, $\varepsilon_2 = \varepsilon_1$, but it is now the sum of an elastic part σ_2/E and a thermal part $\alpha_l \Delta T$. This gives the stress after the heating, σ_2.
3. The condition of the statement, $\sigma_2 = 0$, gives the area, and the area gives the diameter.

This is better than thinking in terms of forces, because the wire does not "know" about the force: it only keeps its length, and the stress is whatever is needed to keep it.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Stress after heating at constant length

Before heating, the strain is $\varepsilon_1 = \sigma_F / E$ with $\sigma_F = F/A$. After heating, the strain is the same, and it is the sum of the elastic and the thermal strain:

$$\varepsilon_2 = \frac{\sigma_2}{E} + \alpha_l \, \Delta T = \varepsilon_1 = \frac{\sigma_F}{E}$$

Solving for the stress after the heating:

$$\sigma_2 = \sigma_F - E \, \alpha_l \, \Delta T$$

The second term is the thermal stress: heating at constant length lowers the tension by $E \alpha_l \Delta T$ (if it were larger than σ_F, σ_2 would be negative, a compression).

2. Condition of the statement

The thermal stress compensates the applied stress when the net stress vanishes, $\sigma_2 = 0$:

$$\frac{F}{A} = E \, \alpha_l \, \Delta T \quad \Rightarrow \quad A = \frac{F}{E \, \alpha_l \, \Delta T}$$

3. Numerical value of the thermal stress

With $E = 207000$ N/mm²:

$$E \, \alpha_l = 207000 \, \frac{\mathrm{N}}{\mathrm{mm}^2} \times 12.5 \times 10^{-6} \, {}^\circ\mathrm{C}^{-1} = 2.5875 \, \frac{\mathrm{N}}{\mathrm{mm}^2 \, {}^\circ\mathrm{C}}$$

$$E \, \alpha_l \, \Delta T = 2.5875 \times 20 \, {}^\circ\mathrm{C} = 51.75 \, \mathrm{N/mm}^2 = 51.75 \, \mathrm{MPa}$$

4. Area and diameter

$$A = \frac{F}{E\alpha_l \Delta T} = \frac{1400\ \text{N}}{51.75\ \text{N/mm}^2} = 27.053\ \text{mm}^2$$

From $A = \pi d^2/4$, $d = \sqrt{4A/\pi}$:

$$4A = 108.21\ \text{mm}^2 \qquad \frac{4A}{\pi} = 34.445\ \text{mm}^2 \qquad d = \sqrt{34.445\ \text{mm}^2} = \mathbf{5.869\ mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $F/(E\alpha_l \Delta T)$: N divided by (N/mm² · °C⁻¹ · °C) = N/(N/mm²) = mm². The product $\alpha_l \Delta T$ has no units, so $E\alpha_l \Delta T$ is a stress.
- **Direct check.** With this area the applied stress is $F/A = 1400/27.053 = 51.75$ MPa, equal to the thermal stress 51.75 MPa, so $\sigma_2 = 0$.
- **Strains.** The thermal strain $\alpha_l \Delta T = 2.500 \times 10^{-4}$ is the same as the elastic strain that the force produced, $\sigma_F/E = 2.500 \times 10^{-4}$: heating gives back exactly the stretch that the force had imposed, and since the length cannot change the stress disappears.
- **Length not needed.** The initial length does not appear in the result. The free expansion of 1 m of wire would be $\alpha_l \Delta T\, l_0 = 0.25$ mm, the amount that the supports prevent.

Interpretation

For a wire thinner than 5.87 mm the applied stress is larger than the thermal one, and some tension remains after the heating. For a thicker wire the applied stress is smaller, and the heating leaves the wire in compression (it could buckle). The result depends on F/A, not on the length, because only the strain matters. As $\Delta T \to 0$ the required area tends to infinity: without temperature rise the force cannot be compensated.

Check against the official solution

Item	Official	Computed	Comment
Diameter	5.86 mm	5.869 mm	agrees (0.15 %): the key is truncated, 5.869 rounds to 5.87